

Introduction to smooth geometry

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Abstract

These are notes for a talk given to the [Boston University Mathematics Association](#) in the Fall 2026 semester. We take a somewhat less rigorous approach to the subject, and focus on breadth and intuition. We introduce manifolds, tangent spaces and the tangent bundle, vector fields the exterior derivative, differential forms, and integration.

Contents

1	Manifolds	1
2	The tangent space	5
3	The tangent bundle	8
4	Differential forms and integration	10

1 Manifolds

We begin with a brief review of calculus. Given a positive integer $n \geq 1$, let $\mathbb{R}^n$ denote the set of all ordered n -tuples $(x_1, \dots, x_n)$, where $x_i \in \mathbb{R}$ for all $1 \leq i \leq n$. For example, an ordered 2-tuple (x_1, x_2) is just an ordered pair. Thus, $\mathbb{R}^2$ is the collection of all ordered pairs. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function which takes an n -tuple $(x_1, \dots, x_n)$, and outputs an m -tuple $f(x_1, \dots, x_n)$.

Calculus 1 and 2 study the calculus of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Given such a function, we can define the *derivative*

$$\frac{df}{dx}.$$

This is a new function $\mathbb{R} \rightarrow \mathbb{R}$ which takes a number $x \in \mathbb{R}$ to the rate of change of f at the point x . However, not all functions take a single input or output. For example, I can write *temperature* as a function $T(x, y, z, t)$ in *four variables*. The variables x, y, z are the coordinates in 3-dimensional space, and t is the time. This outputs a single number: the temperature at the point $(x, y, z) \in \mathbb{R}^3$ at the time $t \in \mathbb{R}$. In calculus, we are very interested in the derivatives and integrals of such a function.

Let's start with a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. For example, consider $f(x, y) = -x^2 - y^2$. I can graph this in 3-dimensionoal space as a sort of surface [draw picture].

For functions $\mathbb{R} \rightarrow \mathbb{R}$, we define the slope at a point as the slope of the line tangent to the curve at that point. However, as we can see in the above image, there are *many* lines we can draw that are tangent to each point. For example, consider the 2 lines I've drawn that are both tangent to f at the point $(0, 0)$: [draw picture].

The issue is that there are now many directions we can go in. In 1 dimension, the only directions we could go in were forward and backward, and clearly going forward on a graph makes more sense than going backwards. But now, we can go left, right, forward, backward, and any direction in-between. So how do we solve this issue?

We start by defining a derivative in a *direction*. If we pick a direction, then we can draw all our tangent lines going in that direction. There are two directions that make a lot of sense: the positive x and positive y directions.

Definition 1.1. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Define the *partial derivative* with respect to x , denoted

$$\frac{\partial f}{\partial x} = \partial_x f,$$

as the derivative of f in the positive x direction. Similarly, the partial derivative with respect to y , denoted

$$\frac{\partial f}{\partial y} = \partial_y f,$$

is the derivative of f in the positive y direction.

These derivatives are nice to calculate. Notice that if we are only travelling in the x -direction, then the y value never changes. In other words, at every point on the line in the below figure, the y value is the same [draw picture]. Thus, the y coordinate is *constant* when we consider the derivative in the x direction.

Thus, to calculate the partial derivative $\partial_x f$, we simply take the normal derivative of f with respect to x , and treat the variable y as a constant. In our example $f(x, y) = -x^2 - y^2$, we have

$$\frac{\partial}{\partial x} (-x^2 - y^2) = \frac{\partial}{\partial x} (-x^2) + \frac{\partial}{\partial x} (-y^2).$$

Since y^2 is a constant, the derivative on the right is the derivative of a constant. Since the derivative of a constant is 0, we have

$$\frac{\partial}{\partial x} (-x^2) + \frac{\partial}{\partial x} (-y^2) = \frac{\partial}{\partial x} (-x^2) + 0 = \frac{d}{dx} (-x^2) = -\frac{d}{dx} (x^2) = -2x.$$

Thus, $\partial_x f = -2x$.

For a more complex example, consider $g(x, y) = 2xy + x^2 + y^3$. We can calculate

$$\frac{\partial g}{\partial x} = \frac{\partial}{\partial x} (2xy + x^2 + y^3) = \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial x} (y^3) = 2y \frac{d}{dx} (x) + \frac{d}{dx} (x^2) = 2y + 2x.$$

We can now consider the question of calculating a total derivative of f , like we were able to for functions in 1 variable. Although we only defined the derivatives of f in the directions x, y , it's not crazy to imagine calculating the derivative of f in an arbitrary direction. Unfortunately, the formulas aren't as nice, which is why we don't start with these derivatives. But this isn't a crazy idea to imagine.

This leads to a natural notion of a "total derivative". Let's go back to our example of $f(x, y) = -x^2 - y^2$. Then $\partial_x f = -2x$. Similarly, we can show $\partial_y f = -2y$. Consider the point $(1, 0) \in \mathbb{R}^2$. Note that $\partial_x f(1, 0) = -2$, but $\partial_y f(1, 0) = 0$. Thus, the derivatives in different directions *can be different* at the same point. A natural idea for a "total derivative" is to consider the derivative such that the derivative is the *biggest*. More precisely, given a point $(x, y) \in \mathbb{R}^2$, we first find the direction such that the derivative of f in that direction is as big as possible, and then we evaluate that derivative at the point (x, y) .

But this means a total derivative has to have two pieces of information: a direction and a magnitude. In other words, a total derivative is a *vector*.

A vector is something that has a direction and a magnitude. We represent it as an arrow that points in the direction it goes, and has the same length as its magnitude. For example, the vector below is an arrow pointing from $(0, 1)$ to $(1, 1)$. It's direction is in the x -direction, and its magnitude (length) is 1. The starting point of a vector is actually not super important. In some cases, it's useful to have a vector start at a particular point in space, but we can almost always figure out what this point is from context. Thus, it's usually easier to forget about the starting point when working with vectors, and only remember the length and magnitude.

This leads to a nice algebraic formulation. By translating all vectors so that they start at $(0, 0) \in \mathbb{R}^2$, we only have to keep track of the endpoint. Thus, if v is a vector starting at $(0, 0)$ and pointing to (x, y) , we simply write $v = (x, y)$.

Definition 1.2. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. The *total derivative* of f , also called the *gradient* of f , is a function $Df : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. It can be calculated as $Df(x, y) = (\partial_x f(x, y), \partial_y f(x, y))$.

Many authors, especially for calculus books, denote the total derivative by ∇f , and call it the gradient. However, we will be going beyond normal calculus, and once we go beyond normal calculus the notation changes to Df , and the name changes to a *total derivative*.

Now consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Then we can write $f(x_1, \dots, x_n)$ as an n -tuple. Using the notation

$$f(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)),$$

we see that f can be built out of a collection of m functions $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$. We define

$$\frac{\partial f}{\partial x_i} = \left(\frac{\partial f_1}{\partial x_i}, \dots, \frac{\partial f_m}{\partial x_i} \right).$$

Thus, the calculus of functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be reduced to the calculus of functions $\mathbb{R}^n \rightarrow \mathbb{R}$. From now on, we focus on these functions.

We will discuss integrals later. For now, I want to move on to the meat of the talk. First of all, note that if I have a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if I write the coordinates of $\mathbb{R}^n$ as $(x_1, \dots, x_n)$, then I can define a partial derivative in any of these directions (and they are calculated the same way). We will often denote the partial of f with respect to x_i as $\partial_i f$. We can then define a total derivative $Df = (\partial_1 f, \dots, \partial_n f)$. However, not everything in real life is $\mathbb{R}^n$.

For example, consider the Earth. The surface of the Earth is a sphere (important note: when a mathematician says “sphere”, they mean a *hollow* sphere. The word “ball” means a *filled in* sphere). Just like the set of real numbers has a symbol $\mathbb{R}$, the sphere has a symbol: S^2 (we will explain the 2 later). If we want to talk about the temperature on the surface of the Earth at a particular time, we would consider a function $f : S^2 \rightarrow \mathbb{R}$ which takes a point $p \in S^2$ to the temperature $f(p)$ at the point p . But how do we do calculus on this function?

Multivariable calculus has an answer for us: we use *spherical coordinates*. Given a point $p \in \mathbb{R}^3$, we can write $p = (r, \theta, \phi)$, where r is the distance from p to the origin, θ is the angle that p makes in the xy -plane, and ϕ is the angle p makes with the z -axis. Some authors flip the usage of θ and ϕ . Regardless of how you define it, you can come up with formulas for derivatives in these coordinates.

But how about something like the torus? How would you define coordinates on this? [draw picture] More precisely, how could you possibly define coordinates that are not extremely messy, such that their derivatives are also not extremely messy?

And what happens when we go to higher dimensions? Even if something doesn’t have a nice geometric interpretation, it can still be important. Let’s say you have a collection of n particles. There are a lot of particles in real life, so n can be very very big. Each particle has 3 position variables, so you now have $3n$ variables if you want to keep track of the position of all particles. Due to restrictions on the particles’ behavior might not be a function from $\mathbb{R}^{3n} \rightarrow \mathbb{R}$, but from some $3n$ -dimensional “geometric space” M to $\mathbb{R}$. How can we reasonably do calculus with such functions?

This is the job of differential geometry. In this talk, we will explore how this calculus can be done. This topic is difficult, so at times we may not be super precise about things, and I may not prove that things work the way I say they do. My goal is to give you a broad overview of this fascinating subject, and hopefully pique your curiosity enough to consider studying it in the future.

The entire field of differential geometry starts with the definition of an n -manifold. The idea is to be precise about what an “ n -dimensional geometric space is”. Additionally, we want this space to support some notion of calculus. Since we already know how to do calculus on $\mathbb{R}^n$, we will use the tools of calculus in the real numbers to build the tools of calculus on manifolds.

First, note that all of calculus so far has happened using coordinates. We could only define the partial derivatives once we had coordinates $(x_1, \dots, x_n)$ for $\mathbb{R}^n$. Using *spherical coordinates*, we can define calculus on the sphere. So if we want to do calculus on a “manifold”, then it should have a notion of coordinates.

Here is how we make that precise. Let M be a *topological space*. This isn’t a topology talk, so we won’t be precise about what a topological space is. All we will say is that you can imagine a topological space as a blob, possibly with holes. [draw picture]

For example, $\mathbb{R}^n$ is a very very straight blob that happens to go on forever, and also happens to have no holes. The sphere S^2 is also a very not-blobby blob, and it has one big hole in the middle of it. The torus is another very simple blob, and it has a big hole in the middle of it. [draw picture]

A continuous function between blobs is just a function that doesn't jump around or tear the blob apart. Intuitively, if p is a point on our blob, then if you zoom in close enough to p , all the points close to p will stay close to p after taking a continuous function.

So a manifold is a blob. However, we want our manifold to have coordinates. The way we say this is as follows.

Definition 1.3. Let M be a topological space (blob). An *open subset* $U \subseteq M$ of M is simply a subset of M that does not contain its boundary. We define *local coordinates on U* to be a function $\varphi : U \rightarrow V$, where $V \subseteq \mathbb{R}^n$ is also an open subset, such that φ is a bijection (meaning it has an inverse), and its inverse is continuous. We write local coordinates as (U, φ) , and will sometimes refer to it as a *chart*, a *coordinate chart*, or simply *coordinates*.

The reason we only consider open subsets is for topological reasons that the reader can safely disregard. The intuition behind definition 1.3 is as follows. Let $U \subseteq M$ be an open subset, and let $p \in U$ be some point. We define the *coordinates of p* to be the point $\varphi(p) \in V \subseteq \mathbb{R}^n$. Since p is a point in $\mathbb{R}^n$, we can write it as an n -tuple $p = (p_1, \dots, p_n)$ in $\mathbb{R}^n$. This is exactly what coordinates are! Coordinates in $\mathbb{R}^n$ are just a way of writing a point as a list of real numbers. By using φ , we can write a point in U as a list of real numbers, hence giving U coordinates.

But why is φ bijective? First, since φ is injective, this means no two different points have the same coordinates. It would be weird if $(p_1, \dots, p_n)$ wasn't a single point, but was instead 2 or more different points. This also means that given any point $(x_1, \dots, x_n) \in V$, this corresponds to a point $\varphi^{-1}(x_1, \dots, x_n) \in U$. In other words, we can think of V as not being a totally different set than U , but rather a copy of U that we use to put coordinates on U . For that reason, we often adopt the following convention.

Given a point $p \in U$, we can write $\varphi(p) = (p_1, \dots, p_n)$. Define the function $x_i : U \rightarrow \mathbb{R}$ such that $x_i(p) = p_i$. Then $\varphi(p) = (x_1(p), \dots, x_n(p))$. We often write $\varphi = (x_1, \dots, x_n)$, and refer to x_i as the *coordinates* of U . Given a point p , we may often write $(x_1(p), \dots, x_n(p))$ simply as $(x_1, \dots, x_n)$ for simplicity, in the same way that we don't write $p = (x(p), y(p))$ when $p \in \mathbb{R}^2$, but instead we write $p = (x, y)$.

Definition 1.4. A *topological n -manifold* is a topological space M with the following property. For every point $p \in M$, there is an open set $U \subseteq M$ such that $p \in U$, and there are local coordinates (φ, U) , where $\varphi : U \rightarrow V$ and $V \subseteq \mathbb{R}^n$ (note that n must be the same number for all of these coordinate charts). In other words, there is always at least one way to write a point in coordinates. This topological space also has to satisfy the following extra conditions, which are only there for topological technicalities, and the reader can safely ignore:

1. M must be second-countable, i.e. it has a countable basis.
2. M must be Hausdorff, i.e. for any points $x \neq y$, there are opens $x \in U$ and $y \in V$ with $U \cap V = \emptyset$.

Some authors also require that manifolds are paracompact.

The idea is that a manifold is built out of patches of $\mathbb{R}^n$. Thus if we zoom in close enough (by taking the open subset U to be really small), we can "flatten out" a patch of our manifold M into a patch of the real numbers $\mathbb{R}^n$. [draw picture]

Finally, we want to be able to do calculus on our manifold. To do this, we can't just use any arbitrary coordinates. We need coordinates that are good for doing calculus. To illustrate what I mean, we must use a bit of circular reasoning. Since this is only motivational, it's totally fine. Once we finish defining manifolds, we will use coordinates to be able to define derivatives of functions $U \rightarrow \mathbb{R}$ for any open subset $U \subseteq M$. We will do this by rewriting our function as a function $\mathbb{R}^n \supseteq V \rightarrow \mathbb{R}$ using coordinates. However, each coordinate x_i is a function $U \rightarrow \mathbb{R}$, so we can consider the derivatives of the coordinates themselves. Now suppose that we choose *bad* coordinates, meaning that the derivatives of x_i don't exist (remember not all functions are differentiable). How are we able to consider derivatives of a function $U \rightarrow \mathbb{R}$ if the coordinates we wrote it in weren't differentiable in the first place?

This is the problem. We need to guarantee that the coordinates we choose are *smooth* in some sense, meaning they have every possible derivative. There is a nifty trick mathematicians have found for ensuring this.

Definition 1.5. Let M be a topological manifold, and let (U, φ) and $(\tilde{U}, \psi)$ be charts, where $\varphi : U \rightarrow V$ and $\psi : \tilde{U} \rightarrow \tilde{V}$. Since φ and ψ are bijective, they have inverses φ^{-1}, ψ^{-1} . Consider the function

$$\varphi \circ \psi^{-1} : V \cap \tilde{V} \rightarrow U \cap \tilde{U} \rightarrow V \cap \tilde{V}.$$

We call this a *transition function*. Since $V, \tilde{V} \subseteq \mathbb{R}^n$, this is a function from a subset of $\mathbb{R}^n$ to a subset of $\mathbb{R}^n$. Thus, we can talk about the partial derivatives of $\varphi \circ \psi^{-1}$. If $\varphi \circ \psi^{-1}$ and $\psi \circ \varphi^{-1}$ are *smooth*, meaning any possible combination of partial derivatives always exists, then we say the charts (U, φ) and $(\tilde{U}, \psi)$ are *smoothly compatible*.

We can now define a smooth manifold.

Definition 1.6. Let M be a topological space. An *atlas* is a collection $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ of coordinate charts of M , such that for all $p \in M$, there is some $i \in I$ with $p \in U_i$. A *smooth atlas* is an atlas where for all $i, j \in I$, the charts (U_i, φ_i) and (U_j, φ_j) are smoothly compatible. A *smooth manifold* is a topological manifold M with a smooth atlas $\mathcal{A}$.

The intuition is that when we work with coordinates on a smooth manifold, we only work with *smooth coordinates*. A coordinate chart (U, φ) is *smooth* if it is smoothly compatible with all of the charts in the smooth atlas.

The idea is that there are many possible smooth coordinates on a manifold, and it would be a hassle to list all of them out. Using an atlas, we only need to list a couple coordinates which define the “smooth structure” of M . For example, we can define a smooth structure for the sphere S^2 as follows. Define

$$U_x^+ = \{(x, y, z) \in S^2 \mid x > 0\}, \quad U_y^+ = \{(x, y, z) \in S^2 \mid y \neq 0\}, \quad U_z^+ = \{(x, y, z) \mid z \neq 0\}.$$

Define the map $\varphi_{x+} : U_x^+ \rightarrow \mathbb{R}^2$ by $\varphi_{x+}(x, y, z) = (y, z)$. This isn't surjective on all of $\mathbb{R}^2$, but by restricting our codomain to the image $\varphi_{x+}(U_x^+)$, we can make it surjective. The trick here is that if $(x, y, z) \in S^2$, then $\sqrt{x^2 + y^2 + z^2} = 1$. Thus, if we know y and z , we can solve

$$x^2 = 1 - y^2 - z^2 \implies x = \pm \sqrt{1 - y^2 - z^2}.$$

Finally, since we know that $x > 0$ in U_x^+ , we know that we need to choose the positive branch of the square root, meaning $\varphi_{x+}^{-1}(y, z) = \sqrt{1 - y^2 - z^2}$. We can do similar stuff for U_y^+ and U_z^+ , and we can also analogously define sets $U_x^-, U_y^-,$ and U_z^- . Our smooth atlas for S^2 is given by these charts $(U_i^\pm, \varphi_{i\pm})$ for $i = x, y, z$.

However, these are not the only coordinates we might want to use on S^2 . For example, *stereographic projection* is another type of smooth coordinates on S^2 , and it's smoothly compatible with these charts. The formula is much more complicated, but it has a nice geometric intuition. If you forget about the north pole of the sphere, then you can stretch out the sphere across the whole plane $\mathbb{R}^2$ [draw picture].

2 The tangent space

So far, we have been very interested in functions $\mathbb{R}^n \rightarrow \mathbb{R}$. Thus, it makes sense to look at functions $M \rightarrow \mathbb{R}$, for M some manifold. To start doing calculus on manifolds, we should start by discussing the notion of a derivative for such a function, since derivatives are historically easiest to study. In calculus, given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, we start by defining the derivative at a *point* $x \in \mathbb{R}$ using the limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

In multivariate calculus, we started by defining the partial derivative $\partial_i f$ of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $x \in \mathbb{R}^n$ using a similar limit definition. Thus, we should start by attempting to define the derivative of a function $f : M \rightarrow \mathbb{R}$ at a point.

Let M be an n -manifold, let $f : M \rightarrow \mathbb{R}$ be any function, and let $p \in M$ be a point. We can use the coordinates of M to obtain a notion of a derivative. Choose a chart (U, x) such that $p \in U$. Recall that this means x is an isomorphism $x : U \cong V \subseteq \mathbb{R}^n$ to some open subset V of $\mathbb{R}^n$. Recall that this allows us to write any point $q \in U$ using coordinates $x(q) = (x^1(q), \dots, x^n(q))$, which we usually just write as $(x^1, \dots, x^n)$ for simplicity. This means that if we only look at the values of f on U , then we can write f as a function in coordinates

$$f(x) : \mathbb{R}^n \rightarrow \mathbb{R}.$$

This should be good enough to define a derivative at p , since the derivative only depends on the value of f in an infinitesimally small region near p .

Definition 2.1. With notation as above, define the *partial derivative of f with respect to x^i at p* to be the value

$$\left. \frac{\partial f(x)}{\partial x^i} \right|_p,$$

where $f(x)$ represents the function f in coordinates, and $|_p$ denotes “evaluation at $x(p)$ ”.

Notation 2.2. We often write $\partial_i|f$ or $\left. \frac{\partial}{\partial x^i} \right|_p f$ to denote the partial derivative of f evaluated at p .

If we want to take multiple partial derivatives of f at p , then we just take the partial derivatives before evaluating at p . For example,

$$\left. \frac{\partial}{\partial x^i} \right|_p \left. \frac{\partial}{\partial x^j} \right|_p f = \left. \frac{\partial^2 f}{\partial x^i \partial x^j} \right|_p.$$

For this reason, we usually only write one evaluation bar. We will usually use notation like

$$\left(\left. \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} \right) \right|_p f \text{ or } \left. \frac{\partial^2}{\partial x^i \partial x^j} \right|_p f.$$

Definition 2.3. Let M be a manifold, and let $U \subseteq M$ be an open subset. Let $f : U \rightarrow \mathbb{R}$ be any function. We say f is *smooth at $p \in M$* if all possible combinations of partial derivatives of f at p exist. The set of all smooth functions $U \rightarrow \mathbb{R}$ is denoted $C_M^\infty(U)$.

If M and N are manifolds, then using the coordinates on both M and N simultaneously, we can provide the same definition for smooth functions $f : M \rightarrow N$.

Although we could work with functions that are not smooth (eg. continuously differentiable functions), smooth functions are much more interesting, are much better behaved, are much more critical to developing the theory, and are still abundant in nature. Thus, we will focus extensively on smooth functions. We will highlight two of these properties right now. If $U \subseteq M$ and $f, g : U \rightarrow \mathbb{R}$ are smooth, then define the functions $f + g, fg : U \rightarrow \mathbb{R}$ by

$$(f + g)(p) = f(p) + g(p), \quad (fg)(p) = f(p)g(p).$$

Then both $f + g$ and fg are smooth functions. In other words, $C_M^\infty(U)$ is a *ring*. For those with knowledge of sheaves, C_M^∞ is a sheaf of rings.

There’s a particular property about partial derivatives at a point p which will help us significantly moving forward: partial derivatives at p form a *vector space*.

Proposition 2.4. *Let M be an n -manifold, and let $p \in M$. Let (U, x) be a chart containing p . Then the partial derivatives $\{\partial_i|_p\}_{1 \leq i \leq n}$ form a basis for a vector space.*

Proof. In order for partial derivatives to form a vector space, we need to know how to add partial derivatives, and how to multiply them by a scalar $c \in \mathbb{R}$. A partial derivative is just a function which takes in smooth functions $C_M^\infty(U)$ (or defined on any subset containing p), and sends them to $\mathbb{R}$. Thus to define $\partial_i|_p + \partial_j|_p$, we just need to know how to evaluate it on a function. Define

$$(\partial_i|_p + \partial_j|_p)f = \partial_i|_p f + \partial_j|_p f.$$

This is a pretty simple definition. The definition for multiplying by a scalar is similarly simple:

$$(c\partial_i|_p)f = c(\partial_i|_p f).$$

The vector space axioms are fairly routine to check, and we omit them for conciseness. The proof that the ∂_i 's are linearly independent is left as an exercise (hint: construct a function $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$ and $p \in \mathbb{R}^n$ such that $\partial_j|_p f^i = 1$ if and only if $i = j$, and 0 otherwise. Using the coordinates $x : U \cong V \subseteq \mathbb{R}^n$, argue that this is sufficient). ■

The vector space constructed in proposition 2.4 is invariant under choice of coordinates. Suppose (V, y) is another set of coordinates, and let φ be the transition function. Recall that this means $y = \varphi(x)$. Let $f : V \rightarrow \mathbb{R}$ be a smooth function, and suppose that I can write f in the coordinates y . Then since $y = \varphi(x)$, we can write f in the coordinates x as well using $f(y) = f(\varphi(x))$. We can now use the chain rule to calculate the partial derivatives of f in the x^i direction:

$$\left. \frac{\partial f}{\partial x^i} \right|_p = \left. \frac{\partial f(\varphi(x))}{\partial x^i} \right|_p = \sum_j \left. \frac{\partial f}{\partial y^j} \right|_p \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p = \left(\sum_j \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p \left. \frac{\partial}{\partial y^j} \right|_p \right) f.$$

Note that the component

$$\left. \frac{\partial \varphi^j}{\partial x^i} \right|_p$$

is just a number, i.e. a *scalar*. Thus,

$$\sum_j \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p \left. \frac{\partial}{\partial y^j} \right|_p$$

is an element of the vector space given by partial derivatives in the coordinates y^i . Thus, we see that the transition function φ gives us a way of transitioning from the vector space of partial derivatives with respect to x^i , to the vector space of partials with respect to y^i . This can also be done in the reverse direction, meaning that these vector spaces are *isomorphic*. In other words, this vector space doesn't depend on the choice of coordinates, since we can always use the transition function to switch between coordinates.

Definition 2.5. The vector space constructed in proposition 2.4 is denoted $T_p M$, and is called the *tangent space to p*.

Now, we can define a total derivative of a function at a point p . Recall that in single variable calculus, the total derivative of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at a point p is just the number $f'(p)$. In multivariable calculus, our best notion of a total derivative was the *gradient vector*, meaning that the total derivative at a point p of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *vector* $\nabla f(p) \in \mathbb{R}^n$. If we have a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we can upgrade the notion of a total derivative to the *Jacobian matrix*: writing $f(x) = (f^1(x), \dots, f^m(x))$, define the Jacobian matrix at p , denoted $Df(p)$, as the matrix of partial derivatives

$$Df(p) = \begin{pmatrix} \partial_1|_p f^1 & \cdots & \partial_n|_p f^1 \\ \vdots & \ddots & \vdots \\ \partial_1|_p f^m & \cdots & \partial_n|_p f^m \end{pmatrix}.$$

Essentially, the derivative of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is an $m \times n$ matrix.

Using this intuition, we can define the derivative of a smooth function $f : M \rightarrow N$ between *manifolds*. Let M be m -dimensional, and let N be n -dimensional. Let $p \in M$, let (U, x) be a chart for p on M , and let (V, y) be a chart for $f(p)$ on N . Then using these coordinates, we can write f as a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. We can then define the *differential of f at p*, denoted df_p , to be the Jacobian matrix

$$df_p = Df(p).$$

This definition has a problem: we want our definitions to be coordinate-independent. That is, if I have different coordinates, I should get the same answer. But that isn't the case here; changing coordinates changes the matrix itself. However, we can fix this by viewing df_p as a *linear transformation*. Recall that a linear transformation can be given by different matrices when we change the basis, and recall that coordinates give us a basis for $T_p M$ and $T_{f(p)} N$. Thus, we will view df_p as a linear transformation $T_p M \rightarrow T_{f(p)} N$, whose *basis representation* is $Df(p)$.

Let's unpack what this means. Let $V \in T_p M$, and use the basis to write

$$V = \sum_i c_i \partial_i|_p, \quad c_i \in \mathbb{R} \forall i.$$

Let's treat the $\partial_i|_p$ as basis vectors, and rewrite this as a column vector:

$$V = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Then we can calculate

$$(df_p)V = \begin{pmatrix} \partial_1|_p f^1 & \cdots & \partial_n|_p f^1 \\ \vdots & \ddots & \vdots \\ \partial_1|_p f^m & \cdots & \partial_n|_p f^m \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} c_1 \partial_1|_p f^1 + \cdots + c_n \partial_n|_p f^1 \\ \vdots \\ c_1 \partial_1|_p f^m + \cdots + c_n \partial_n|_p f^m \end{pmatrix}.$$

If we rewrite this column vector using the basis ∂_j (here we take the partials wrt y), we have

$$= \sum_i \sum_j c_j \frac{\partial f^i}{\partial x^j} \Big|_p \frac{\partial}{\partial y^i} \Big|_p.$$

Using this formula, we can calculate that

$$(df_p)(\partial_j|_p) = \sum_i \frac{\partial f^i}{\partial x^j} \Big|_p \frac{\partial}{\partial y^i} \Big|_p.$$

Let's look at the special case of a function $f : M \rightarrow \mathbb{R}$. Then we note that $T_p \mathbb{R}$ always has a basis given by $\partial_x|_p$, since $\mathbb{R}$ has a global set of coordinates given by normal coordinates (we call this a *global chart*). This means that we can write

$$(df_p)(\partial_i|_p) = \frac{\partial f}{\partial x^i} \Big|_p \frac{\partial}{\partial x} \Big|_p.$$

We will come back to this calculation. For now, we will close with the following remark about tangent spaces.

If f is a function $f : \mathbb{R} \rightarrow \mathbb{R}$, then one definition of $f'(x)$ is the *slope of the line tangent to f at the point x* . If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then one definition of ∇f is the *tangent vector to the graph of f with maximal slope*. Continuing this trend, there is a way of thinking about df as a vector tangent to f . In particular, notice that $\partial_i|_p$ produces tangent vectors, and thus we can think of $\partial_i|_p$ as a tangent vector *itself*. The vector $\partial_i|_p$ is the vector tangent to p which points in the x^i -direction. This geometric interpretation of the algebraic object $\partial_i|_p$ turns out to be well-motivated; there is a theorem proving that this interpretation works. In this way, we think of the entire tangent space $T_p M$ as the space of vectors that are tangent to p . This interpretation will be useful.

3 The tangent bundle

I don't know about you guys, but I'm tired of carrying around the "evaluation at $p \in M$ " symbol. If we have coordinates on an open set U , we can do derivatives anywhere on U using those coordinates, so why do we have to carry around a symbol for derivatives at a single point? Additionally, we have been working extensively with coordinates. But coordinates can be messy, and are not necessarily defined everywhere. It would be nice to have some coordinate-independent properties of derivatives. The *tangent bundle* TM will solve all of these problems.

The idea of the tangent bundle is exactly what the name says: we bundle together all of the tangent spaces into a new manifold. More precisely, the definition is as follows.

Definition 3.1. Let M be an n -manifold. Define the *tangent bundle* TM to be the manifold

$$TM = \{(p, v) \mid p \in M, v \in T_p M\}.$$

The explicit description of the smooth structure is not important, but we make two comments. First, TM is a $2n$ -manifold. Second, there is a function $\pi : TM \rightarrow M$ given by $\pi(p, v) = p$. This map is smooth.

The tangent bundle is the prototypical example of a vector bundle. A *vector bundle* is a fascinating object which allows us to bridge algebraic geometry and smooth geometry through the language of sheaves, and I recommend the interested reader to look into them after this paper. For now, we will focus on the tangent bundle.

The power of the tangent bundle comes from the notion of a *section*.

Definition 3.2. Let M be a manifold, and let $\pi : TM \rightarrow M$ be the projection of the tangent bundle. A *smooth section* s of the tangent bundle TM over an open set $U \subseteq M$ is a smooth function $s : U \rightarrow TM$ such that for all $p \in U$, we have $\pi(s(p)) = p$. The set of all smooth sections over U is denoted $\mathfrak{X}(U)$.

There is another term for a section of TM over U : a *vector field*. To motivate this terminology, we investigate what it means for $\pi(s(p)) = p$. First, note that we must have $s(p) \in \pi^{-1}(p)$. Note that

$$\pi^{-1}(p) := \{(p, v) \mid \pi(p, v) = p\} = \{(p, v) \mid v \in T_p M\} = \{p\} \times T_p M.$$

Thus, $s(p)$ must be of the form (p, v) , where $v \in T_p M$. We can identify $s(p)$ with the component v (since there is an isomorphism $\{p\} \times T_p M \cong T_p M$). Thus, a section s is simply a choice of a vector $s(p)$ which is tangent to p for all $p \in U$. It is convention to write s_p instead of $s(p)$ for sections, and we adopt this convention in what follows.

Now imagine drawing a manifold, and at each point, drawing a vector which goes out from that point, and is tangent to the curves of the manifold. This is exactly what a vector field is.

We can now finally drop the evaluation at p symbol from partial derivatives using the following proposition.

Proposition 3.3. Let x^i be coordinates on some open set $U \subseteq M$. Then for all i , there is a vector field $\partial_i \in \mathfrak{X}(U)$ given by the assignment $(\partial_i)_p = \partial_i|_p$.

Proof. We will not worry too much about smoothness, so we will simply say that once we have explicitly defined the smooth structure on TM , the smoothness of this proposition is obvious. The definition itself is mildly complex however, hence why we have omitted it. As for showing that $\pi((\partial_i)_p) = p$, we simply note that $(\partial_i)_p = \partial_i|_p \in T_p M \cong \pi^{-1}(p)$, and thus the statement follows. ■

This statement has a useful corollary.

Corollary 3.4. Let $X \in \mathfrak{X}(U)$ be a vector field, and let x^i be coordinates on U . Then there are smooth functions $X^i : U \rightarrow \mathbb{R}$ such that for all $p \in U$,

$$X_p = \sum_i X^i(p) \left. \frac{\partial}{\partial x^i} \right|_p.$$

Proof. Let $p \in U$. Then $X_p \in T_p M$. As noted before, the collection $\{\partial_i|_p\}_i$ forms a basis for $T_p M$, and thus we can write

$$X_p = \sum_i c_i \partial_i|_p$$

for some coefficients $c_i \in \mathbb{R}$. We simply define $X^i(p) := c_i$. Using the smooth structure, it can be proven that these functions are smooth. ■

Thus, given a vector field X , every component vector X_p is a differential operator. This suggests the following definition.

Definition 3.5. Let $f : U \rightarrow \mathbb{R}$ be a smooth function, and $X \in \mathfrak{X}(U)$ a vector field. We define the *action* of X on f , denoted Xf , to be the *smooth function*

$$Xf : U \rightarrow \mathbb{R}$$

$$(Xf)(p) = X_p f.$$

Here, the notation $X_p f$ indicates that we are writing X_p as a differential operator, and then applying that differential operator to f . Since this differential operator lives in $T_p M$, it already includes evaluation at p , so we do not need to write something like $(X_p f)(p)$.

There is a similar definition, where we act $C^\infty(U)$ on $\mathfrak{X}(U)$:

Definition 3.6. Let $f : U \rightarrow \mathbb{R}$ be a smooth function, and $X \in \mathfrak{X}(U)$ a vector field. Define the *action* of f on X also called the *multiplication* of X and f , to be the vector field $fX \in \mathfrak{X}(U)$ given by

$$(fX)_p = f(p)X_p.$$

Note that since $f : U \rightarrow \mathbb{R}$, we have $f(p) \in \mathbb{R}$. Since X_p is a vector in a vector space $T_p M$, and since vector spaces are closed under scalar multiplication, scaling X_p by the scalar $f(p)$ simply yields another vector in $T_p M$. Thus, this is indeed a vector field.

We can now elevate most of the properties of derivatives to the level of vector fields. For example, the chain rule holds at this level. The multivariate chain rule (especially in coordinates) can be a bit messy to work with without using computers, so we will look at a simpler rule: the product rule (also called the Leibniz rule). Given a vector field X and smooth maps $f, g : U \rightarrow \mathbb{R}$, for any $p \in U$, we have

$$X(fg)(p) = (Xf)(p) \cdot g(p) + f(p) \cdot (Xg)(p).$$

This can be proven using coordinates, however this has the benefit of *not requiring coordinates at all* in order to be used. Just because the proof uses coordinates doesn't mean that we have to choose coordinates in order to use this fact. Thus, this fact can apply anywhere on a manifold, not just in a small area where we happen to have coordinates. This is the power of "coordinate-independence".

Before closing, we also note that $\mathfrak{X}(U)$ is a vector space, with $(X + Y)_p := X_p + Y_p$.

4 Differential forms and integration

Now, we want to look at the differential df of a function f . Note that this is *not a vector field*: although for all p we have a function df_p , this function is not a differential operator. Instead, it is a sort of function of differential operators. Thus, df cannot live in $\mathfrak{X}(U)$. We will need to create a new vector space for df to live in. Luckily, we know exactly what space this should be.

Recall that if $f : M \rightarrow N$, then df_p was a linear map $T_p M \rightarrow T_{f(p)} N$. Thus, given a vector field X , we can define $df(X) = (df_p)(X_p) \in T_{f(p)} N$, and thus $(df)(X) \in \mathfrak{X}(f(M))$. Therefore, $df \in \text{Hom}_{\mathbb{R}}(\mathfrak{X}(M), \mathfrak{X}(f(M)))$.

This is a sort of messy definition, and we can simplify it by specializing to our most interesting case: where $f : U \rightarrow \mathbb{R}$, for some open subset $U \subseteq \mathbb{R}^n$. In fact, we can gain a lot of useful intuition by specializing even further. Let x^i be coordinates on U . Then each individual x^i is a function $\mathbb{R}^n \rightarrow \mathbb{R}$. What is dx^i ?

Consider dx^i for x^i some coordinate. Show that $dx^i(\partial_j) = \delta_{ij}$ under the identification $T_p \mathbb{R}^n \cong \mathbb{R}^n$. Introduce T^*M as the dual bundle, and define the pullback map. Introduce df as a section $\Gamma(-, T^*M) =: \Omega^1$. Write $d : C^\infty(U) \rightarrow \Omega^1(U)$. Discuss $\mathbb{R}$ -linearity and $d(fg) = (df)g + fdg$. Write a general 1-form as $(fdx^i)_p = f(p)dx^i|_p$ in coordinates. Argue that dx^i is indeed a differential in the direction x^i : a derivative is the operator giving us infinitesimal changes, and so this is like the length of that operator, which is infinitesimal... idk. Consider a line integral

$$\int_\gamma f dx^i = \int_0^1 \gamma^*(f dx^i) = \int_0^1 f(\gamma(t)) \gamma'(t) dt.$$

Finally, argue using orientations that we should work in $\Gamma(-\wedge^\bullet T^*M)$. Define

$$d\omega = \sum_i \frac{\partial \omega_I}{\partial x^i} dx^i \wedge dx^I.$$

Argue $d^2 = 0$. Note that this yields the de Rham complex, and note the fascinating quasi-isomorphism to the singular cochain complex, and homotopy-invariance of $\Omega^\bullet$. Given a top form ω , define

$$\int_U \omega.$$

Note that some manifolds (Riemannian manifolds) have a canonical top form, called vol. . We then define the integral of $f : M \rightarrow \mathbb{R}$ to be

$$\int_U f \text{vol.}$$

If M doesn't have a canonical top form, then it makes less sense to think about forms as measures, and more sense to think about them as the different things you integrate (since the interesting thing probably is the diff measures u get). Write Stokes' theorem

$$\int_M d\omega = \int_{\partial M} \omega.$$

Use this to prove the fundamental theorem of calculus. Note that this subsumes the fundamental theorem of line integrals, Stokes' theorem for vector fields, and Gauss' theorem. If time, remark on sheaves to motivate AG.